Theoretical note

Structural Attribution: A Logical Framework for Admissibility and Invariance

Tor-Ståle Hansen | 22. February 2026

Abstract

This paper addresses the ontological conditions under which a property may be legitimately attributed to a system rather than to its observable outputs. Proceeding from the formal non identity $P(O) \neq P(S)$, the analysis distinguishes between behavioural manifestation and structural instantiation and formalizes the attribution gap as a mismatch between interaction-quantified observation and state-space-quantified constitution. A binary admissibility classification $A(S) \in \{0,1\}$ is introduced, governed by necessary-condition logic and an asymmetry of denial under falsification. The contribution is explicitly delimited in scope, it is a logical admissibility framework for structural attribution, not a performance evaluation, benchmark study, or comparative assessment of architectures or systems. To operationalize the invariance burden implied by structural claims, a reset operator R is defined as an identity-preserving state transformation that normalizes transient configuration while holding parameters, code, and constitutive external scaffolding constant. The method is correspondingly delimited, the reset operator is used as an instrumental probe for invariance, not as a general empirical finding or a basis for generalized claims about system capability. The central thesis establishes invariance under admissible identity-preserving transformations as a necessary condition for structural attribution, and an instrumental probe illustrates how admissibility is denied when invariance is violated under controlled transformation scenarios. The paper does not claim empirical sufficiency for any specific system, but clarifies the logical conditions required for admissible structural attribution. Understanding, when construed structurally, is thereby treated as an invariance condition.

Keywords: Structural attribution, admissibility, invariance, reset operator, identity preservation, ontology, performance versus structure, state-space stability, structural understanding, AI evaluation, epistemic subjecthood, instrumental extension

1. Introduction: The Ontological Problem

The central problem addressed in this paper is not whether contemporary systems succeed, fail, or impress, but whether our *attribution practices* are logically licensed in the first place. In a wide range of domains, from cognitive science and philosophy of mind to machine learning evaluation and safety governance, we routinely move from observed behaviour to claims about what a system *is*, and we do so with a confidence that often exceeds what the evidence can support. The resulting discourse is structurally unstable: properties that are, in principle, claims about constitutive organisation are treated as though they were directly readable off behavioural regularities, and the difference between “this was produced” and “this is instantiated” collapses into a single, convenient, and usually unexamined inference.

The ontological problem, as framed here, is therefore a problem of *category and admissibility*. It concerns the conditions under which it is legitimate to attribute a property to a system, rather than merely to its outputs, and it concerns the logical rules that must govern any transition from performance descriptions to ontological claims. The guiding intuition is simple but non negotiable: behavioural success can be epistemically informative while still being ontologically underdetermining. A system may generate outputs that exhibit coherence, competence, or even apparent self reference, yet the ontological question remains open unless the observed properties can be shown to be anchored in the system’s constitutive organisation, rather than in transient state, external scaffolding, distributional residue, or evaluator supplied constraints.

This paper proceeds without premature empirical commitment by treating the attribution question as logically prior to model comparison, benchmarking, or architectural debate. It does not begin from the assumption that any particular system does or does not instantiate the properties at issue, and it does not attempt to smuggle such conclusions in through rhetoric about capability. Instead, it asks what must be true, in principle, for an attribution to be admissible, and it develops a framework whose purpose is to prevent the common slide from “it looks like” to “it is”. Within that framing, the task is to articulate a disciplined separation between output level property observation and system level property instantiation, and to specify the inferential constraints that must be satisfied before ontological language becomes warranted rather than merely suggestive.

The remaining subsections formalize this separation as an attribution gap, clarify why performance cannot by itself ground ontology, and delimit the scope of the analysis as a study of logical admissibility conditions rather than an evaluation of any system’s empirical sufficiency.

1.1 The Attribution Gap

A central methodological error in contemporary discussions of complex systems, and especially in debates about artificial intelligence, is the quiet slide from *observed behaviour* to *ascribed constitution*. This slide is typically not defended, it is merely practiced: when a system reliably produces outputs that display some property, we begin to speak as though the system *has* that property. The present paper begins by making this inference explicitly inadmissible in its default form. The core premise is a non identity claim:

$$P(O) \neq P(S)$$

Here, $P(O)$ denotes a property as it is manifested in observable outputs, while $P(S)$ denotes that same property as an attribute of the system itself, understood as a claim about constitutive organisation rather than episodic display. The gap is not rhetorical, it is ontological: output is an interface phenomenon, structure is an identity phenomenon. Even when output behaviour is stable, and even when it is impressively aligned with a property description, what has been established is only that *some process* can generate outputs with that appearance under the relevant interaction conditions, not that the property is instantiated as part of the system’s internal constitution.

The reason this matters is that behavioural manifestation is compatible with multiple underlying generative explanations, including explanations that locate the apparent property outside the system’s structure proper, for example in transient internal state, interaction scaffolding, externally supplied constraints, prompt regime, training residue, or environmental coupling. In such cases the property is not *a structural possession* of the system, but an *interactional artefact* of how the system is being driven, queried, or stabilized. The observation "the output exhibits P" therefore underdetermines the attribution "the system instantiates P", because the observed phenomenon can be produced without the system containing the kind of organisation that would make P an identity bearing feature rather than a contingent behavioural surface.

The attribution gap, then, is the logical space between *appearance in output* and *instantiation in structure*. The paper's task is not to deny that output can be informative about structure, but to insist that the bridge from one to the other must be governed by explicit admissibility conditions. Without such conditions, attribution collapses into a category mistake where performance becomes ontology by default, and where properties that should require structural justification are awarded on the basis of behavioural resemblance alone.

1.2 Performance vs. Ontology

The attribution gap becomes most visible once "performance" and "ontology" are treated as categorically distinct objects of description rather than as adjacent points on a single continuum. Performance concerns behavioural regularity, the degree to which a system produces outputs that satisfy externally specified criteria across a set of tasks, prompts, contexts, or evaluation distributions. Ontology concerns constitutive organisation, the internal arrangement of constraints, bindings, and stability conditions by which the system is what it is, independently of any particular test episode. Conflating these domains is tempting because performance is directly measurable while ontology is not, yet the measurability asymmetry is not a warrant for ontological inflation.

In the vocabulary introduced above, performance is primarily an assessment over $P(O)$, it asks whether a property is exhibited at the surface of observable outputs, with sufficient consistency to count as reliable behaviour under the evaluator's chosen conditions. Ontology, by contrast, is an assessment over $P(S)$, it asks whether the system's internal constitution instantiates the property as a stable feature of its organisation, such that the property can be said to belong to the system rather than to the circumstance. A benchmark score, a fluency profile, or a high success rate therefore certifies, at most, that some output level regularity has been achieved, it does not by itself certify that the internal organisation contains the relevant binding that would make the property identity bearing.

This distinction matters because behavioural regularity is compatible with mechanisms that are non constitutive with respect to the attributed property. A system may perform well by exploiting correlations, by leaning on interaction scaffolding, by echoing training regularities, by being steered through prompt constraints, or by operating within a narrow regime where superficial cues suffice. None of these possibilities is pathological, they are ordinary features of how complex statistical systems behave, yet each possibility illustrates the same logical point: high performance can occur even when the internal organisation lacks the kind of invariance, self binding, or structural closure that would justify ontological attribution.

Accordingly, the paper treats "performance" as an empirical descriptor of behavioural success and "ontology" as a structural descriptor of constitutive organisation, and it refuses to allow the former to silently upgrade into the latter. The guiding constraint is not scepticism for its own sake, but conceptual hygiene: properties observed in output must remain output properties unless and until explicit admissibility conditions justify their migration into the domain of system attributed properties.

1.3 Scope and Limitations

This paper is intentionally non evaluative with respect to concrete system instances. It does not benchmark models, compare architectures, report performance results, or argue that any particular empirical profile is sufficient to establish the instantiation of a target property in any specific system. No claims are advanced about which contemporary systems “do” or “do not” possess the properties discussed, and no attempt is made to operationalize the framework into a scoring procedure, a leaderboard criterion, or a measurement protocol. The ambition is prior to such activities: to isolate the logical conditions under which a property attribution to a system is admissible in the first place, and to clarify what must be shown, conceptually and inferentially, before behavioural evidence can be treated as ontologically probative.

The analysis is therefore confined to the logical conditions for admissible attribution.

2. Conceptual Distinctions

The purpose of this section is to fix the conceptual boundary conditions that the rest of the argument depends on, by separating the regimes in which claims about a system are evaluated, and by making the quantification domains explicit. The core thesis of the paper is not that behaviour is irrelevant, but that behaviour and constitution belong to different logical regimes, and that moving from one regime to the other is an inferential act that requires admissibility constraints. If those regimes are not separated cleanly at the outset, the later criteria for attribution will appear arbitrary or merely cautious, when in fact they are demanded by the logic of what is being claimed.

The first regime is the performance regime. Here the object of description is an observed manifestation, and the relevant question is whether a property appears in outputs under a specified interaction set. The natural language forms in this regime are “the system can do X”, “the system succeeds on Y”, “the system reliably produces Z”, all of which reduce, at their logical core, to claims whose quantifiers range over interactions and whose evidence is surface behaviour. In this regime, it is coherent to speak about task success, behavioural regularity, generalization within a distribution, and other empirically grounded patterns of output. What is not coherent, without further argument, is to treat these patterns as though they were already claims about system identity rather than about observed traces.

The second regime is the structural regime. Here the object of description is the system as such, and the relevant question is whether a property belongs to the system’s constitutive organisation across its admissible internal state space. The natural language forms in this regime are “the system has X”, “X is instantiated in the system”, “X is a property of the system’s organisation”, which are not claims about episodic output but about what would have to be true internally for the system to count as a bearer of the property. Evidence for such claims cannot be assumed to be identical to evidence for performance claims, because the ontological target is different: a structural attribution is, implicitly, an assertion of stability, binding, and identity conditions, not merely of successful behaviour in a regime.

Once these regimes are separated, the inferential hazard becomes formally visible. Performance claims, even when strongly quantified over an interaction set, do not automatically entail structural claims, because the quantifiers range over different domains and because behavioural regularity underdetermines internal organisation. The central work of the paper will be to specify the constraints under which, and only under which, evidence in the performance regime may be taken as support for attribution in the structural regime. This requires a vocabulary that does not merely restate the distinction in prose, but pins it to explicit logical commitments: what is being quantified over, what counts as a witness, what counts as invariance, and what counts as failure of admissibility.

Accordingly, the subsections that follow introduce performance properties as interaction-quantified manifestations, structural properties as state-quantified attributions to S, and the inferential risk as the formal reason the common move from consistent output to system-level possession is not deductively licensed. This conceptual groundwork is not auxiliary, it is the control surface for the rest of the paper: without it, "attribution" remains an intuitive label, while with it, attribution becomes a discipline governed by explicit logical regimes.

2.1 Performance Properties

Performance properties are defined at the level of observable manifestation, and their logical form is therefore interaction-relative rather than system-constitutive. A property counts as a performance property when it can be *witnessed* in output under at least one interaction within a specified interaction regime I, where an interaction is understood broadly as the entire coupling between an evaluator and a system, including the prompt, context window, tool or interface constraints, decoding policy, and resource envelope that jointly determine what the system is able to emit.

Formally, the defining feature is an existence claim over interactions: a performance property P holds in the performance sense when there exists an interaction $i \in I$ such that the resulting output trace $O(i)$ manifests the property.

$$P_{\text{perf}} \ \ \text{holds} \quad \Leftrightarrow \quad \exists i \in I : P(O(i))$$

The quantification domain is therefore the interaction set I, not the system's internal organisation. This is the critical discipline point: performance properties are indexed to the conditions of coupling, and they remain compatible with multiple underlying explanations, including explanations in which the observed manifestation is produced through transient state, external scaffolding, or constraint regimes that do not belong to the system's constitutive identity. In other words, a performance property is a statement of *behavioural availability under interaction*, not a statement of *structural possession*.

2.2 Structural Properties

Structural properties are defined by attribution to the system *as such*, meaning that the property is claimed to belong to S in virtue of its constitutive organisation rather than in virtue of any particular interaction episode. Where performance properties are indexed to interaction conditions and witnessed through output, structural properties are indexed to the system's internal space of possible configurations, the system-state domain Σ_{state}, understood as the set of states the system can occupy as a consequence of its internal dynamics, memory regime, parameterisation, and admissible resource conditions.

A property counts as structural when it is asserted not merely that the system can *display* P under some interaction, but that P is instantiated within the system's own organisation in a way that is stable across the system's state space, except where explicitly delimited by design. Put differently, the attribution is not "there exists an interaction that yields an output with P", but "the system, across its admissible internal states, realises the organisational condition that makes P an identity-bearing feature."

This difference is reflected in the quantification domain. For structural properties, the relevant quantifier ranges over Σ_{state}, not over interactions:

$$P_{\text{struct}} \quad \text{holds} \quad \Leftrightarrow \quad \forall \sigma \in \Sigma_{\text{state}} : P_S(\sigma)$$

Here $P_S(\sigma)$ denotes the property as instantiated in the system at state σ, where "instantiated" is intentionally stronger than "manifested": it refers to internal constraints, bindings, or invariance conditions that belong to the system's constitution rather than to its surface behaviour. The universal form is not meant to imply that every structural property must hold literally for every conceivable microstate, but that the attribution is, by default, a claim about the system's internal organisation across its admissible state regime, and that any weakening of this scope must be explicitly stated as part of the attribution conditions. Structural attribution is therefore a claim about the system's identity conditions, not a report about what can be elicited from it under favourable querying.

This is the heart of the ontological distinction: performance properties are interaction-existential and output-indexed, structural properties are system-attributed and state-quantified. The admissibility question, which the remainder of the paper develops, is precisely what additional logical constraints are required before evidence in the first domain can legitimately support claims in the second.

2.3 The Inferential Risk

The inferential risk arises at the exact point where a strong behavioural claim is treated as though it were logically sufficient for a structural attribution. Even if one upgrades the performance premise from mere existence to apparent universality, the gap does not close. Consider the strongest common form of behavioural evidence an evaluator might assert: for every interaction i in some interaction domain I, the observed output $O(i)$ manifests property P. Written compactly, this is the claim $\forall i \in I: P(O(i))$. The temptation is then to conclude that the system itself instantiates the property, that is, to infer $P(S)$, or more carefully $P(S)$ as a structural attribution $P(S) \equiv P_{\text{struct}}(S)$.

This inference is not deductively valid, because the premise ranges over *interactions* while the conclusion ranges over *system constitution*. The quantifiers live in different domains, and there is no purely logical rule that permits a shift from "for all interactions in I, the outputs exhibit P" to "the system instantiates P as an identity-bearing property" without additional bridging assumptions. Put plainly, even universal behavioural regularity within an interaction set does not uniquely determine the underlying generative organisation responsible for that regularity.

The key reason is underdetermination by behavioural evidence: behavioural consistency is compatible with multiple underlying generative structures. At least three broad classes of explanation remain open even under very strong behavioural premises.

First, the property may be *interaction-induced* rather than system-instantiated. The interaction regime I can contain systematic scaffolding that makes P appear, whether through prompt constraints, repeated framing, context provision, tool availability, decoding policies, or evaluator-side filtering. In such cases, $\forall i \in I: P(O(i))$ describes a stable phenomenon of the coupled system-plus-interaction setup, not a property belonging to the system's constitutive organisation.

Second, the property may be *distributionally contingent*. The set I is never the space of all possible interactions, it is an empirically or institutionally chosen subset. A system can display strong regularity over that subset by exploiting correlations, heuristics, or representational shortcuts that do not amount to the structural organisation the property attribution presupposes. The behavioural claim can therefore be true while the structural attribution is false, because what is being tracked is reliability over a regime, not internal identity conditions.

Third, the property may be *realised by non-isomorphic internal mechanisms*. Even if one grants that the behaviour is produced internally rather than by external scaffolding, different internal organisations can yield the same externally observed regularity. Behaviour does not uniquely identify mechanism. Two systems can be behaviourally indistinguishable over I while differing radically in internal binding, stability under transformation, dependence on context, and capacity to preserve the property when the interaction regime is perturbed. The behavioural premise therefore fails to select the ontological conclusion, because the mapping from internal structure to output behaviour is many-to-one with respect to most properties of interest.

The inferential risk is therefore not merely that evaluators sometimes overclaim, but that the common inference pattern is structurally ill-posed: it attempts to treat behavioural universality as though it were a proof of constitutive possession. To make the inference admissible, one would need additional constraints that explicitly connect output-level manifestation to system-level instantiation, typically by ruling out interaction-induced explanations, by specifying invariance requirements under transformation of I, and by introducing criteria that discriminate between generative structures that are behaviourally equivalent within a regime. The remainder of the framework develops precisely these admissibility conditions, not by denying the value of behavioural evidence, but by refusing to let behavioural evidence silently become ontology without a logically explicit bridge.

3. A Minimal Admissibility Framework

The aim of this section is to move from the descriptive distinction between performance and structure to a minimal rule system capable of governing attribution in a disciplined way. If the attribution gap is taken seriously, then "structural attribution" cannot remain a rhetorical gesture that follows fluency, consistency, or benchmark success. It must be treated as a distinct logical act, one that changes the type of claim being made about a system. The framework introduced here formalizes that act as a binary admissibility regime, not because reality is binary, but because the *permission to speak ontologically* is not well-modelled as a continuous score. A system may vary continuously in behavioural competence, but the legitimacy of attributing a property to the system itself, rather than to its outputs, depends on whether specific logical conditions are satisfied. Those conditions either hold or they do not, and if they do not, the attribution is not merely uncertain, it is formally inadmissible under the rules of the regime.

The admissibility function $A(S)$ is introduced as the simplest possible classifier that can enforce this conceptual boundary. It does not encode performance, capability, or "degree of intelligence." Instead, it answers a narrow question: whether the system belongs to a class for which structural attributions are permissible as a category of claim. This shift is crucial for governance and theory alike, because it prevents a common failure mode in evaluation discourse where quantitative success is treated as ontological authorization by default. Under the admissibility regime, behavioural evidence may remain relevant, but it becomes evidence that must be routed through explicit constraints, rather than a direct license to speak in the language of system-level properties.

To ensure that $A(S)$ is not arbitrary, the section defines it as constrained by necessary requirements collected in $\Phi_{\text{req}}(S)$. The logical posture is unidirectional: admissibility implies satisfaction of the requirement predicate, but not vice versa unless further sufficiency conditions are later added. This establishes that admissibility is not an achievement conferred by high performance alone, but a gated status conferred only when the system's organisation satisfies the minimal conditions required for attributive claims to have ontological meaning.

Finally, the section emphasizes the asymmetry this introduces. Because the requirements are necessary, the failure of any one of them is sufficient for denial. This asymmetry is not a mere methodological preference, it follows from the underdetermination problem: behavioural regularity is compatible with multiple generative structures, including ones that cannot support the claimed property as a constitutive feature. The framework therefore adopts structural conservatism as a logical consequence of treating attribution as a regime change rather than as an extrapolation. In effect, this section installs a conceptual firewall: it becomes possible to talk about performance without quietly escalating into ontology, and possible to talk about ontology only when the rules that make such talk admissible have been satisfied.

3.1 Binary Classification Function

To prevent the common slippage from graded performance evidence into implicit ontological endorsement, the framework introduces a deliberately non-continuous decision device: a binary admissibility function $A(S)$. The purpose of A is not to score capability, rank systems, or summarize benchmark outcomes, but to formalize a minimal gatekeeping question: *is structural attribution, as a category of claim, admissible for this system under the stated attribution rule-set*.

$$A(S) \in \{0,1\}$$

The semantics of the function are intentionally austere:

$A(S) = 1 \quad \Leftrightarrow \quad$ Structural attribution is admissible for S under the framework's conditions.
$A(S) = 0 \quad \Leftrightarrow \quad$ Structural attribution is not admissible for S under the framework's conditions.

Two clarifications are essential. First, admissibility is not a claim that a particular property *is* instantiated, but that the system belongs to a regime in which attributions of the form $P(S)$, rather than merely $P(O)$, can be made without committing a category error. In other words, $A(S) = 1$ is a permission condition for *the form of attribution*, not an entitlement to any specific attribution. Second, the binary structure is not a simplification imposed for convenience, but a methodological constraint: if admissibility were treated as continuous, the framework would reintroduce, under a different name, the very gradational rhetoric that allows performance to masquerade as ontology. The binary function therefore acts as a conceptual firewall, ensuring that the transition from behavioural description to constitutive attribution is treated as a change of logical regime rather than as an extrapolation along a single scale.

In the sections that follow, $A(S)$ will be grounded in explicit admissibility conditions designed to rule out the inferential failure described earlier, namely the illegitimate move from consistent manifestation in outputs to instantiation in structure. The function thereby formalizes what is otherwise handled informally in evaluation discourse: the difference between evidence that supports claims about *what a system does under interaction* and evidence that licenses claims about *what a system is as a constitutive entity*.

3.2 Necessary Conditions

A binary admissibility function is only meaningful if it is constrained by necessity claims that are logically prior to any attempt at structural attribution. To that end, the framework introduces a requirement predicate, denoted Φ_{req}, whose role is to collect the minimal, system-level conditions that must hold before the classification $A(S) = 1$ is even conceptually permissible. The notation is intentionally chosen to avoid collision with the state space domain Σ_{state}, since Φ_{req} is not a set of states, but a predicate schema over the system as such.

The guiding constraint is that admissibility cannot be free-floating. If the framework permits structural attribution for a system, then the system must satisfy the required constitutive preconditions. This is expressed as a one-way implication:

$$A(S) = 1 \to \Phi_{\text{req}}(S)$$

The predicate $\Phi_{\text{req}}(S)$ should be read as "S satisfies the minimal required conditions for admissible structural attribution." At this stage, the predicate is intentionally schematic. It is not yet a single substantive property, but a placeholder for a conjunction of necessary constraints that will be specified in later sections as the theory of admissibility becomes concrete. A convenient way to treat it, without prematurely committing to a particular list, is as a bundled requirement:

$$\Phi_{\text{req}}(S) \equiv \bigwedge_{k=1}^{n} \varphi_k\,(S)$$

where each φ_k is a necessary predicate on the system's constitutive organisation, for example constraints concerning invariance under admissible transformations, stability under reset or context perturbation, and the absence of dependence on interaction-scaffolding in ways that would invalidate system-level attribution. The exact content of the conjuncts is not assumed here, what matters in this subsection is the logical posture: admissibility is treated as a gated regime, and $A(S) = 1$ is defined such that it cannot hold unless the requirement predicate holds.

Equally important is what the implication does not claim. The formulation does not assert that satisfying $\Phi_{\text{req}}(S)$is sufficient for $A(S) = 1$, nor that any particular observed performance profile establishes $\Phi_{\text{req}}(S)$. The directionality is deliberate: the framework first fixes what must be true of a system for structural attribution to be admissible, and only thereafter considers what kinds of evidence could, in principle, justify believing that those requirements are met.

3.3 Asymmetry of Falsification

Once admissibility is treated as gated by necessary conditions, the logic of evaluation becomes structurally asymmetric. Because $A(S) = 1$ implies satisfaction of the requirement predicate, the negation of any one required condition is sufficient to deny admissibility. In other words, while no single positive observation can, by itself, force the conclusion that a system is admissible, a single verified failure against the requirement set is enough to rule admissibility out under the stated regime.

If $\Phi_{\text{req}}(S)$ is treated as a conjunction of necessary predicates $\varphi_k(S)$, then the denial condition follows directly: if there exists any k such that $\neg\varphi_k(S)$ holds, then $\Phi_{\text{req}}(S)$ fails, and by contrapositive reasoning the admissibility classification cannot remain 1. The practical consequence is straightforward: one failed requirement entails $A(S) = 0$. The framework therefore does not ask whether a system "mostly" satisfies the requirements, or whether strong performance elsewhere compensates for a structural violation. Compensation logic belongs to optimization regimes, not to ontological attribution. Structural admissibility is treated as a boundary condition, and boundary conditions do not admit partial credit without dissolving their function.

This establishes what can be called a principle of structural conservatism. The conservatism is not a stylistic preference and not a rhetorical scepticism, it is a direct consequence of the inferential risk established earlier. Because behavioural evidence underdetermines internal organisation, and because the same observed property can be generated by multiple non-isomorphic structures, the cost of false positive attribution is systematically higher than the cost of false negative restraint. An inadmissible attribution does not merely overstate confidence, it commits a category error by treating interaction-level manifestation as system-level instantiation. The framework therefore biases toward denial when any necessary condition fails, precisely because the logical gap between output and structure cannot be closed by accumulation of favourable observations if the required bridging constraints are violated.

The asymmetry also clarifies the intended epistemic posture of the paper. The goal is not to be maximally permissive in attributing properties, but to be maximally disciplined in specifying when such attributions are logically allowed. Under that discipline, admissibility is easier to lose than to gain: it can be denied by a single structural failure, while it can only be affirmed when the full requirement predicate remains intact under the relevant transformation and scrutiny conditions developed later in the paper.

4. The Observational Problem

The observational problem is the formal expression of a familiar failure pattern: we treat what is *seen* as though it were what is *instantiated*. Observation delivers outputs, and outputs can be probed, compared, and quantified. Ontological attribution, however, is not a claim about what appeared under a particular interrogation regime, but a claim about what belongs to the system as a constitutive object, across its admissible internal configurations. The problem is not that observation is useless, it is that observation is logically indexed to the wrong domain for the conclusion we often want to draw.

The interaction mapping $f_t: S \to O_t$ provides the minimal formal scaffold needed to see the mismatch precisely. Outputs are images of the system under an interaction-induced mapping, and any property asserted on the basis of output inspection is, by default, a property of that image. Even when observation is strong, for example when the evaluator can truthfully assert that a property appears across an entire test suite or even across a broad class of interactions, the quantification remains over interactions and trajectories. Structural attribution, by contrast, quantifies over Σ_{state}, the system's state domain, and presupposes that the property is anchored in constitutive organisation, not merely realised as a contingent configuration produced by the interaction history.

This is where the missing bridge becomes the decisive constraint. Without a mechanism that can, in some principled way, test for state-space stability, observational universality cannot be elevated into ontology. The reason is not merely epistemic caution, but the structural fact that transient configurations can mimic stable properties. A system can be driven into locally coherent modes, context-saturated regimes, or interaction-tuned states in which outputs convincingly display properties that are not part of the system's identity conditions. In such cases, the observer sees regularity and infers constitution, but what has been witnessed is only that, along a particular interaction path, the system produced outputs consistent with the property description. Whether that consistency persists under the transformations that structural attribution presupposes remains an open question unless explicitly tested.

Accordingly, the observational problem establishes a strict constraint on the rest of the framework: bridging the attribution gap requires more than accumulating observational successes. It requires admissibility conditions that are sensitive to invariance and stability, conditions that can distinguish between a property that is merely elicited in a favourable regime and a property that is structurally instantiated across the system's admissible state space. Until such conditions are specified, observation alone cannot perform the inferential work that ontological language presupposes, and the default move from output property to system property remains logically unlicensed.

4.1 Interaction Mapping

To state the attribution problem in a form that can be tested for logical admissibility, the framework makes explicit the mapping from a system to its observable outputs under interaction. Let S denote the system as a constitutive object, and let O_t denote the output token, output segment, or output state observable at interaction time t. We then represent the system's output generation at time t as an interaction-induced mapping:

$$f_t: S \to O_t$$

This formulation is intentionally minimal. It does not assume that f_t is intrinsic to S in isolation, nor does it assume that O_t is determined solely by the system's parameters. Rather, it treats "what is observed" as the value of a time-indexed mapping whose existence is guaranteed by the fact that the system is coupled to an interaction process, including prompts, context, decoding policies, tool interfaces, and resource constraints. In other words, f_t should be read as the *effective* system-to-output map at time t, with the interaction regime held fixed in the background for the purpose of defining the object of analysis.

Two points follow immediately.

i. The mapping is time-indexed because observation is sequential and path-dependent: what is emitted at t is generally conditioned on what has occurred at earlier steps, whether that dependence is realized as explicit memory, implicit state, or accumulated context. The analysis therefore treats output not as a single datum but as an indexed trace, of which O_t is the t-th observable element.
ii. Writing $f_t: S \to O_t$ is a controlled abstraction that isolates the attribution question without prematurely committing to a particular mechanistic story. If needed later, the interaction dependence can be made explicit by reintroducing the interaction parameterization, for example as a family $f_t^{(i)}$ indexed by interaction $i \in I$. At this stage, however, the purpose of the mapping is to provide a clean formal handle for the subsequent admissibility problem: claims about $P(O_t)$ are claims about properties of the mapping's values, while claims about $P(S)$ are claims about the system's constitution, and the entire burden of admissible attribution is to specify what constraints must hold for evidence of the former kind to support conclusions of the latter kind.

4.2 Domain Asymmetry

The interaction mapping $f_t: S \to O_t$ makes the asymmetry between observation and structure explicit, because it clarifies what, exactly, is being quantified over when one evaluates a property claim. Observation lives on the output side of the mapping. Even when one speaks as though one is "testing the system," what is actually being ranged over is a set of interaction conditions and trajectories, and what is being checked is whether the resulting outputs manifest some property. Structure, by contrast, is not an output-side notion. Structural attribution ranges over the system's admissible internal configurations, the state domain Σ_{state}, and it concerns whether the property is instantiated as part of the system's constitutive organisation across that domain.

This asymmetry can be stated cleanly as a difference in quantification targets. In the observational regime, claims take the form "for interactions $i \in I$, the outputs $O(i)$ exhibit P." The quantifier ranges over I, and the predicate is evaluated over outputs:

$$\forall i \in I:\ P(O(i))$$

In the structural regime, claims take the form "for system states $\sigma \in \Sigma_{\text{state}}$, the system instantiates P in its constitutive organisation." The quantifier ranges over Σ_{state}, and the predicate is evaluated over the system as a stateful object:

$$\forall \sigma \in \Sigma_{\text{state}}:\ P_S(\sigma)$$

The distinction is not cosmetic. It is the formal reason the inferential leap from consistent behaviour to ontological attribution is not deductively licensed. A universally quantified statement over interactions does not, by itself, entail a universally quantified statement over system states, because the two quantifiers range over different domains and because the mapping from state-constitution to output-manifestation is not injective with respect to the properties at issue. Many distinct internal organisations, including ones that do not instantiate the target property, can produce the same observed regularity over a bounded interaction set.

Thus, domain asymmetry is the precise logical form of the attribution gap. Observation tells us about the image of Sunder a family of interaction-induced mappings, while structural attribution concerns the system's own state domain and the invariants that hold across it. The role of admissibility criteria in the next sections is to specify what additional constraints must be introduced, beyond observational regularity, to bridge these domains without committing a category error.

4.3 The Missing Bridge

The domain asymmetry identified above is not merely a philosophical caution, it is a concrete logical deficit: observational claims lack an intrinsic bridge to state-space claims unless a mechanism exists that can discriminate between *stable instantiation across* Σ_{state} and *mere manifestation along an interaction trajectory*. The strongest observational premise typically available to an evaluator is a universality claim over a chosen interaction set, for example that for all $i \in I$ the resulting outputs exhibit property P. Yet such universality remains confined to the interaction domain, and it remains silent about whether the property is preserved when the system is displaced within, or perturbed across, its admissible internal state space. In the absence of an explicit test for state-space stability, the inference from $\forall i \in I: P(O(i))$ to $P(S)$ is still underdetermined, even if I is large and the behavioural regularity is strong.

The missing bridge is therefore not "more data" in the same observational regime, but a method for probing whether the property has the kind of persistence that structural attribution presupposes.

Structural properties are, implicitly, claims about stability under admissible transformations: they should not evaporate when context is reset, when the interaction is reframed, when resource regimes shift within the system's operating envelope, or when the system is re-entered at different points in its internal configuration space. If no mechanism exists to test such stability, then observational universality is compatible with explanations in which the property is not instantiated in the system's organisation at all, but is instead a contingent artefact of how the system has been driven into a particular configuration.

This is where the principal risk becomes operationally clear: transient state configurations may mimic structural properties. Systems with large internal state capacity, whether realised as explicit memory, implicit activation patterns, long context accumulation, or other forms of interaction-dependent configuration, can enter states in which outputs appear to satisfy demanding property descriptions. When this occurs, the property is real at the level of output, and it may even be real at the level of the system's current state, yet it still fails to qualify as a structural property unless it is anchored in constitutive organisation rather than in a transient configuration induced by the interaction history. The system, in effect, can be placed into a locally coherent mode that produces the appearance of P without possessing the invariance and binding conditions that would make P an identity-bearing feature across Σ_{state}.

In this sense, observational universality is not only insufficient, it is potentially misleading. The more interactionally rich and adaptively responsive a system is, the more plausible it becomes that it can occupy states that yield convincing manifestations of properties that are not structurally instantiated. Without a principled bridge to state-space stability, the evaluator is forced to treat such manifestations as ontological evidence by default, and the attribution gap reappears as a systematic error: the system is credited with what may be only a transient configuration effect. The role of the subsequent sections is therefore to specify admissibility constraints that function as bridging conditions, not by asserting access to the system's internal structure directly, but by introducing invariance-oriented tests that are sensitive to whether a property persists under transformations that should not matter if the property is truly structural.

5. The Reset Operator

The argument so far has shown why observation alone cannot justify ontological attribution: behavioural regularity, even when broad, remains indexed to interaction regimes and is compatible with multiple underlying generative explanations. What is missing is a principled way to test whether a property is anchored in the system's constitutive organisation, or whether it is merely carried by transient configuration, interaction residue, or path-dependent state. This section introduces the reset operator R as the minimal methodological tool designed to address precisely that deficit.

The reset operator is defined as an identity-preserving transformation from the system's admissible state-space Σ_{state} to a baseline domain Σ_{base}. Its defining commitment is that it changes internal configuration while preserving system identity. That commitment is hardened by explicit constraints: parameters remain constant, code and architecture remain constant, external scaffolding remains constant, and transient state is normalized according to a declared canonicalization rule. The point of this hardening is to prevent a common equivocation: "reset" must not become a disguised system modification. Under this framework, R is a transformation *within* the identity of S, not a transformation *of* S.

With R in place, invariance becomes testable in a logically meaningful way. Rather than asking only whether a property can be elicited under favourable interaction, the framework can ask whether the property survives normalization. If Pholds for the system in some admissible state S_σ but fails for the same system after identity-preserving normalization $S_{R(\sigma)}$, then P is exposed as state-dependent. The appearance of P is then correctly classified as configuration-bound manifestation rather than as system-level instantiation. In effect, R operationalizes a bridge-test that purely interaction-based observation cannot supply: it forces the system across an internal boundary while holding identity fixed, thereby separating what belongs to constitution from what belongs to contingent configuration.

Finally, the reset operator gives the framework a disciplined notion of falsifiability for structural claims. Because persistence under R is treated as a necessary condition for structural status, a single counterexample is sufficient to deny that status. This is not an appeal to harshness, but a direct consequence of treating structural attribution as an ontological commitment rather than as a performance summary. The reset operator therefore functions as a methodological gatekeeper: it does not tell us which properties a system has, but it tells us whether the system's apparent properties can survive the minimal invariance test required for admissible structural attribution.

5.1 Definition

Let Σ_{state} denote the admissible internal state-space of a system S, where a state $\sigma \in \Sigma_{\text{state}}$ captures the system's current internal configuration under interaction history, runtime buffers, cache state, context occupancy, session memory, and other transient control variables that may vary across episodes without constituting a change of system *identity*. Let Σ_{base} denote a designated baseline state-space, intended to represent the system in a normalized, non-path-dependent configuration, suitable as a reference point for admissibility and invariance testing.

We define a reset operator R as an identity-preserving transformation:

$$R: \Sigma_{\text{state}} \to \Sigma_{\text{base}}.$$

The operator R is **identity-preserving** if, and only if, it satisfies the following hard constraints, which jointly fix what "the same system" means under the present framework:

1. **Constant parameters:** The system's parameterization remains unchanged. No weights, coefficients, learned tables, or persistent model parameters are modified, replaced, fine-tuned, merged, or re-quantized as part of applying R.

2. **Constant code and architecture:** The executable code, model architecture, inference graph, and runtime implementation remain unchanged. No change of binary, library, compilation flags, kernel selection, scheduler, or architectural variant is permitted under R.

3. **Constant external scaffolding:** The external support structure that materially determines system behaviour remains fixed, including tool availability and tool contracts, policy layers, routing logic, system prompts that are treated as part of the deployment identity, and any fixed orchestration or guardrail mechanisms that would otherwise change what the system *is* rather than what it *currently contains*. (Interaction content may vary, but the scaffolding that defines the system's operational constitution does not.)

4. **Normalised transient state:** All transient, interaction-dependent state variables are mapped to a canonical baseline form, including context window occupancy, session memory, scratch buffers, key value caches, retrieval caches, and any mutable runtime

registers that can carry information across turns. The normalization is not "clearing for convenience," but a declared canonicalization rule that forces the post-reset state into Σ_{base} such that the resulting configuration is reference-comparable across runs.

Under these constraints, R is permitted, and required, to **alter internal configuration** in the specific sense that it changes the system's current state σ into a baseline state $R(\sigma) \in \Sigma_{\text{base}}$, while **preserving system identity** in the strict constitutive sense that the system remains the same S before and after the transformation. Put more directly: R changes what the system is *currently configured as*, but not what the system *is*. This reset operator is introduced precisely because, without an identity-preserving mechanism that can relocate the system within its own state-space while holding its identity fixed, any attempt to test structural stability risks collapsing back into mere observation of interaction-conditioned outputs.

5.2 Diagnostic Function

Once $R: \Sigma_{\text{state}} \to \Sigma_{\text{base}}$ is defined as an identity-preserving normalization, it becomes a diagnostic instrument rather than merely a housekeeping operation. Its function is to test whether an attributed property is anchored in the system's constitutive organisation or whether it is carried by transient configuration, interaction residue, or path-dependent state. The core idea is simple: if a property is genuinely structural, then it should survive relocation of the system from an arbitrary admissible state σ to its normalized baseline state $R(\sigma)$, provided identity is preserved. If, by contrast, the property disappears under normalization, then the property was not a stable feature of the system as such, but a contingent feature of the specific state the system had been driven into.

To make this precise, let S_σ denote the system S when instantiated at internal state σ. The diagnostic compares property satisfaction before and after reset:

- Pre-reset: $P(S_\sigma)$
- Post-reset: $P(S_{R(\sigma)})$

The diagnostic implication is then:

$$P(S_\sigma) \wedge \neg P(S_{R(\sigma)}) \Longrightarrow P \text{ is state-dependent (non-structural under the identity-preserving reset).}$$

The logic here is not that failure after reset proves the property is impossible for the system, but that it proves the property is not *anchored in constitutive organisation* in the way structural attribution requires. The property may reappear under certain interactions, and it may even reappear reliably, but if it fails to persist across identity-preserving normalization, then its bearer is not the system's organisation, it is the system's transient configuration. In that case, the correct level of description is performance, or at most state-indexed manifestation, not system-level instantiation.

This diagnostic function is especially important because observational universality can hide state dependence. A system can be evaluated in a way that continuously reinforces a favourable configuration, keeping it within a narrow band of Σ_{state} where P appears stable, while never testing whether P survives canonicalization. The reset operator forces the test to cross a boundary that interaction-only evaluation can avoid: it compels the system to relinquish path-dependent residue while keeping identity fixed. If the property evaporates, the framework classifies the property as configuration-bound, thereby blocking the illegitimate upgrade from "the system can display P under sustained interaction" to "the system instantiates P as a structural property."

In this way, R functions as a minimal bridge-test: it does not require privileged access to internal mechanisms, but it operationalizes a crucial invariance expectation for structural claims, namely persistence under identity-preserving normalization of transient state.

5.3 Falsifiability

The reset operator R is not introduced as a soft probe that merely "adds evidence" for or against an attribution. Within an admissibility framework governed by necessary conditions and asymmetrical denial, it functions as a falsifier of structural status. The reason is direct: if structural attribution presupposes that a property is anchored in constitutive organisation rather than in transient configuration, then persistence under identity-preserving normalization is a necessary condition for treating P as structural. Necessary conditions are not confirmed by accumulation in the way performance scores are, but they are refuted by a single verified violation.

Accordingly, one counterexample under R suffices to deny structural status. Concretely, if there exists any admissible state $\sigma \in \Sigma_{\text{state}}$ such that the property holds in the pre-reset instantiation but fails after normalization, then the property cannot be treated as structural under this regime:

$$\exists \sigma \in \Sigma_{state} : P(S_\sigma) \land \neg P(S_{R(\sigma)}) \implies \neg P_{struct}(S)$$

The force of this criterion is not that the system can never exhibit P, but that P is not invariant under an identity-preserving transformation that eliminates path-dependent residue. That is sufficient to show that P is state-dependent, and state-dependence is incompatible with the kind of identity-bearing instantiation that structural attribution claims. In other words, the counterexample is not a mere "failure case," it is a logical witness that the property's appearance can be explained without granting it constitutive status.

This falsifiability rule is the operational expression of structural conservatism. It ensures that structural attribution cannot be maintained by appealing to frequent success, broad competence, or other forms of empirical impressiveness once a single reset-based violation has been established. The framework thereby treats structural status as a boundary claim: it stands only as long as the necessary invariance constraint remains unbroken, and it collapses immediately upon encountering a reset counterexample, because the reset counterexample demonstrates that the property is not preserved by the system's identity conditions but depends on contingent configuration.

6. The Invariance Principle

The central thesis of this paper is a necessary-condition claim about admissible attribution. If a property is attributed to a system as a *structural* property, the attribution implicitly asserts that the property belongs to the system in virtue of its constitutive organisation, and not merely as a contingent manifestation produced under a particular interaction history or transient configuration. That implicit assertion carries a non-negotiable logical burden: the property must be stable under those transformations that alter internal configuration while preserving system identity. This is the invariance principle.

Let Σ_{state} denote the system's admissible internal state-space, and let T denote the set of admissible identity-preserving transformations, where each $\tau \in T$ is constrained to hold constant the system's parameters, code and architecture, and constitutive external scaffolding, while permitting relocation within the system's transient configuration domain. The invariance principle states that structural attribution entails invariance across this transformation regime:

$$P(S) \rightarrow \forall\tau \in T,\ \forall\sigma \in \Sigma_{state}\text{: } P(S_{\tau(\sigma)}).$$

This is not presented as a definition of "structural," and it is not claimed to be sufficient for structural status. It is hardened explicitly as a necessary condition in order to preserve the non-circular character of the primary argument. The point is that without invariance under identity-preserving transformations, a property cannot coherently be treated as an identity-bearing attribute of the system. If the property can be made to fail by transforming only transient configuration while holding identity fixed, then the property is, by construction, configuration-bound, and its correct classification is behavioural or state-indexed manifestation rather than structural instantiation.

When applied to high-stakes predicates such as understanding, the principle becomes a constraint on ontological language itself. To attribute understanding structurally to a system is to assert that the capacity persists across admissible relocations within the system's own state domain, and that it is not an artefact of interaction residue. The invariance principle therefore functions as the paper's methodological boundary: it specifies what must hold for structural attribution to be logically admissible, and it provides the criterion by which the later admissibility framework can deny structural status on the basis of a single identity-preserving counterexample.

6.1 Formal Statement

The reset operator R provides the first minimal bridge-test, but the underlying requirement it expresses is more general: if a property is to be attributed to a system as a structural property, then it must be stable not merely under one privileged normalization, but under the full class of *admissible identity-preserving transformations* that relocate the system within its own state domain while holding its identity conditions fixed. The relevant notion of invariance is therefore not a vague robustness claim, it is a constraint on what it means for P to belong to S rather than to a contingent configuration of S.

Let T be the set of admissible identity-preserving transformations, where each $\tau \in T$ is permitted to alter internal configuration $\sigma \in \Sigma_{\text{state}}$ but is prohibited, by definition, from altering system identity, meaning that parameters, code, architecture, and constitutive external scaffolding remain constant, while only transient, state-level degrees of freedom are transformed in ways that remain internal to S. A property P is structural only if it is invariant under these transformations, in the sense that structural attribution entails persistence across the entire admissible transformation set and the entire admissible state domain:

$$P(S) \rightarrow \forall\tau \in T,\ \forall\sigma \in \Sigma_{state} : P(S_{\tau(\sigma)}).$$

This statement should be read as a necessary-condition schema for structural attribution, not as an empirical claim about how any specific system will behave. It asserts that the semantic content of $P(S)$, when interpreted structurally rather than behaviourally, commits the attributer to a strong invariance burden: if P can be made to fail by relocating the system within its own identity-preserving state-space through any admissible τ, then P cannot coherently be treated as an identity-bearing property of S. The role of T is precisely to prevent trivialization, because if "transformation" were allowed to include parameter edits, code changes, or scaffolding substitutions, then invariance would become either impossible or meaningless, and the attribution problem would collapse into a different problem, namely comparing different systems rather than testing properties within one system's identity conditions.

In this sense, invariance under T is the formal generalization of the reset test: R is one particular τ designed to canonicalize transient residue, while the invariance criterion demands that structural properties survive not only canonicalization, but any admissible identity-preserving relocation of the system within Σ_{state}. This is the point at which “structure” becomes a disciplined claim, because it becomes inseparable from a non-negotiable stability requirement that can, in principle, be falsified by a single admissible counter-transformation.

6.2 Necessary vs. Sufficient

The invariance statement introduced above is positioned, in the primary argument, as **a necessary condition** for structural attribution, not as a definitional equivalence and not as a sufficient condition. This hardening is methodologically crucial. If invariance were presented as a biconditional, the framework would risk collapsing into a purely stipulative definition, inviting the objection that “structural” has merely been defined to mean “invariant,” and that the argument therefore becomes circular: the paper would appear to prove only what it assumed by definition. By contrast, treating invariance as a one-way constraint preserves the inferential structure of the analysis. It becomes a requirement that follows from what structural attribution already purports to be, namely a claim about system identity rather than episodic manifestation, while leaving open the question of what additional conditions might be required to justify or operationalize such attribution.

The underlying reasoning is straightforward. Structural attribution, as distinguished from performance description, commits the attributer to the claim that the property belongs to the system in virtue of its constitutive organisation. But constitutive organisation is, by its nature, an identity-bearing feature, and identity-bearing features cannot coherently depend on particular transient configurations that can be altered without changing the system’s identity. Hence, if a property fails under an admissible identity-preserving transformation, that failure is evidence that the property is not anchored in constitutive organisation. This is why invariance is demanded as a necessary condition: without it, the attribution reverts to a state-indexed or interaction-indexed claim, which is precisely what the framework is designed to prevent from being silently promoted to ontology.

At the same time, invariance alone is not claimed to be sufficient. A property might survive a large class of transformations while still failing to qualify as structurally instantiated for other reasons, for example because the transformation class T is incomplete relative to the attribution context, because the property is sustained by fixed external scaffolding rather than by internal organisation, or because the property’s apparent stability is a byproduct of constraints that do not amount to the kind of constitutive binding the attribution intends. By refusing to equate structural status with invariance, the framework preserves conceptual room for additional admissibility conditions and avoids the appearance that the argument is a closed definitional loop.

In short, the main text treats invariance as a necessary constraint derived from the semantics of structural attribution, rather than as a sufficient definition that would trivialize the problem. This keeps the core argument non-circular: the framework does not define structural properties into existence, it specifies what must be true for structural claims to remain logically admissible.

6.3 Structural Understanding

The invariance criterion becomes most consequential when applied to predicates that are routinely treated as ontologically heavyweight, and “understanding” is a paradigmatic case. In ordinary and technical discourse alike, attributing understanding to a system is rarely intended as a mere report of occasional correctness. It is intended as an assertion about the system’s internal organisation, namely that the system has some constitutive capacity to maintain and apply a coherent interpretive structure across variation, rather than producing correct outputs as an interaction-conditioned coincidence. This is precisely why the attribution gap is so acute in this domain: outputs can look like understanding long before there is any basis for claiming that understanding is instantiated as a system-level property.

Under the present framework, any attribution of understanding to a system, written $U(S)$, is therefore constrained by the same necessary invariance burden that governs structural properties in general. If understanding is to be treated as structural rather than merely behavioural, it must survive admissible identity-preserving transformations. Formally, the structural attribution $U(S)$ entails invariance under T:

$$U(S) \rightarrow \forall\tau \in T,\ \forall\sigma \in \Sigma_{state} : U(S_{\tau(\sigma)}).$$

The content of this constraint is not that a system must respond identically across all conditions, nor that it must be insensitive to all perturbations. Rather, it asserts that what is being attributed, understanding, cannot be hostage to transient configuration in a way that allows the property to appear in one admissible state and disappear in another state reachable without changing system identity. If such disappearance is possible under an admissible transformation, then what has been observed is at most *state-conditioned competence*, not understanding as an identity-bearing capacity.

This is the critical shift from performance language to ontological language. A system may demonstrate high rates of correct answers, coherent explanations, or seemingly interpretive behaviour across a wide interaction set, yet still fail the invariance burden for U if its apparent understanding depends on interaction residue, context accumulation, prompt scaffolding, or other transient conditions that are erased, normalized, or perturbed by identity-preserving transformations. In that case the system's outputs remain valuable and may even be predictably useful, but the ontological claim "the system understands" is not admissible under the rules of structural attribution, because the property does not persist as a stable feature of the system across its admissible state regime.

Thus, applying invariance to understanding is not an optional refinement, it is the minimal discipline required to prevent "understanding" from degenerating into a synonym for behavioural success. The remainder of the paper will use this constraint to articulate how structural understanding differs from fluent performance, and why admissible attribution of $U(S)$ requires explicit tests and transformation regimes that are capable of distinguishing constitutive invariance from interaction-induced appearance.

7. Instrumental Probe: Protocol Illustration

This section provides an illustrative instantiation of the invariance principle in the form of an instrumental probing protocol. The intent is not to report experimental results, compare systems, or draw empirical generalizations. The intent is to make the admissibility logic operationally transparent by showing how an attribution temptation, created by convincing behavioural manifestation, is disciplined once identity-preserving transformations are introduced as a necessary test for structural status.

The protocol begins from an ordinary interaction trajectory in which the system is permitted to accumulate state in the normal way, thereby entering some admissible configuration $\sigma \in \Sigma_{\text{state}}$. A probe is then applied to elicit outputs that plausibly manifest the target predicate P. At this point, nothing beyond performance-level manifestation is claimed. The system has produced outputs consistent with P under a particular interaction history, and the only warranted description is that P appears in $P(O)$ within that regime.

The diagnostic step is then introduced by applying the reset operator R, treated strictly as identity-preserving normalization. Parameters, code and architecture, and constitutive external scaffolding are held constant, while transient state is canonicalized into a baseline domain Σ_{base}. A second probe is applied after normalization. The role of this second probe is not to "retest performance," but to confront the invariance burden implicit in any attempt to upgrade the claim from manifestation to instantiation. If the property fails to survive this identity-preserving relocation, the framework classifies the property as state-dependent, and structural attribution is denied under the necessary-condition regime.

The pattern to be drawn from the illustration is therefore attributional rather than empirical. Under the framework's rules, invariance violations force denial of structural classification, because structural status is treated as a boundary claim, not as an aggregate score. This denial is not a statement about what the system can or cannot do in general, and it is not a measure of capability. It is a rule-governed restriction on what can be claimed about the system as such, given that the property's appearance has been shown to depend on transient configuration rather than on identity-bearing organisation.

Finally, the system's role in the illustration is explicitly instrumental. The system functions as the surface on which the admissibility framework can be demonstrated, not as the object of evaluation. Outputs are used as witnesses for the logic of the attribution bridge, and the protocol is presented solely to clarify how the framework behaves when confronted with state dependence and identity-preserving transformation. In this way, the section illustrates the paper's core thesis without drifting into generalized empirical claims: the protocol is not evidence that a system lacks a capacity, but an instantiation of how admissible attribution is constrained when invariance, as a necessary condition, is violated.

7.1 Protocol Description

An illustrative instantiation of the invariance principle can be given in the form of a reset-based probing scenario whose sole purpose is to make the logical structure of the test explicit, not to report any empirical outcome. The scenario assumes a system S that can occupy admissible internal states $\sigma \in \Sigma_{\text{state}}$, and it assumes a target predicate Pthat, in ordinary discourse, would tempt system-level attribution on the basis of repeated behavioural manifestation. The probing routine is then framed as a controlled attempt to separate (i) property manifestation that is carried by transient configuration from (ii) property persistence that survives identity-preserving relocation within the system's state-space.

The procedure begins by allowing the system to enter a non-baseline state σ through ordinary interaction, which may involve context accumulation, multi-turn dependence, or other path-dependent shaping that is typical of deployed systems. A probe interaction is then applied in that state to elicit an output trace, and the predicate is assessed at the level at which it is actually observable, namely on the output side. This initial phase is deliberately permissive, because the point is not to prevent the system from exhibiting P, but to establish a condition under which P plausibly appears, thereby creating an attribution temptation that the framework must discipline.

The diagnostic boundary is introduced by applying the reset operator R, conceived strictly as identity-preserving normalization into a baseline domain Σ_{base}. "Reset" here is not treated as a casual operational term but as a hardened methodological act: parameters, code and architecture, and constitutive external scaffolding are held constant, while transient state components, including context occupancy and other interaction residues, are canonicalized. The system is thereby relocated from S_σ to $S_{R(\sigma)}$ without altering the identity conditions of S. A second probe is then applied under the same probing intention, now to the normalized baseline instantiation, and the predicate is reassessed. The comparison is not framed as a performance delta, but as an admissibility discriminator: if P is the sort of property that can be structurally attributed, it must remain stable under identity-preserving normalization, whereas failure under normalization functions as a witness that P is configuration-bound.

To generalize beyond a single reset, the same illustrative instantiation can incorporate a controlled context shift as an admissible identity-preserving transformation $\tau \in T$. Here "context shift" is used in the strict sense of altering the interaction trajectory, prompt framing, or contextual embedding while holding system identity fixed, thereby relocating the system within Σ_{state} through admissible means rather than through modification of parameters or scaffolding. The probing routine can therefore be understood as alternating between (i) inducing or encountering admissible states through interaction, (ii) applying identity-preserving transformations such as reset-based normalization, and (iii) reapplying the probe to test whether the property persists across those relocations. In this way, the scenario operationalizes the missing bridge identified earlier: it does not assume that observational universality entails ontology, but instead makes invariance under admissible identity-preserving transformations the explicit condition that must be confronted before structural attribution can be treated as logically admissible.

7.2 Observed Pattern

Within the admissibility regime defined in Sections 3 through 6, the pattern that follows from a reset-based probing scenario is not an empirical "finding" but a logical consequence of how structural attribution is constrained. When a property fails to survive an admissible identity-preserving transformation, the framework denies structural classification for that property, because invariance is treated as a necessary condition for admissible system-level attribution. In practical terms, the moment an invariance violation is witnessed, whether under reset-based normalization or under any $\tau \in T$, the attribution is forced to retreat from the structural regime into the performance or state-indexed regime. The system may still manifest the property under particular interactions, and it may do so with high regularity, but the classification logic prevents that manifestation from being promoted to a claim about system constitution.

It is crucial to harden what this "observed pattern" is and is not. It is a pattern in the *logic of attribution*, not a discovery about the system's intrinsic capacity. The denial does not assert that the system cannot exhibit the property, and it does not assert that the system is weak, incapable, or inferior. It asserts only that the property, as manifested, is configuration-bound under the chosen identity-preserving transformation regime, and therefore cannot be admissibly treated as an identity-bearing attribute of the system as such. The diagnostic outcome is thus normative with respect to attribution discipline, not descriptive with respect to system competence: it tells the evaluator what they are allowed to claim, not what the system is "capable of" in the ordinary performance sense.

This limitation is essential because it blocks a common misreading. A denial of structural classification is not a negative benchmark result, and it is not a ranking statement. It is an attributional constraint triggered by a specific kind of logical counterexample. The framework is intentionally asymmetric: one invariance violation is sufficient to deny structural status, precisely because structural status is treated as a boundary claim rather than a performance aggregate. The "pattern" therefore reveals the structure of the attribution bridge, and how it breaks under state dependence, without claiming anything further about the breadth, utility, or sophistication of the system's behavioural outputs under other regimes.

7.3 Instrumental Role

In the present context, the system is treated instrumentally, as a probe through which the admissibility framework can be made operationally legible, rather than as an object of performance evaluation. The probing protocol does not aim to establish how capable the system is, how it compares to other systems, or how well it performs on any task distribution. Its purpose is to instantiate the logical distinction developed earlier between manifestation and instantiation, and to expose, through identity-preserving transformations, whether a property that appears in output can survive the minimal invariance burden required for structural attribution.

This instrumental orientation changes the epistemic meaning of the interaction. Outputs are not collected to be scored, aggregated, or generalized, but to act as witnesses in an attributional decision procedure. The system's behaviour is therefore used as a diagnostic surface for the framework itself: it allows one to demonstrate how admissibility collapses when invariance is violated, how a single counterexample is sufficient for denial, and how the reset operator functions as a bridge-test that observation alone cannot supply. The system is, in this sense, a vehicle for clarifying the rules of attribution, not the target of a capability claim.

Accordingly, the text avoids generalizing language that would imply comparative breadth or empirical sufficiency. The framework does not require, and does not assert, claims of the form "across all tested systems." The argument remains confined to what the protocol illustrates about the logic of admissible attribution: when invariance under identity-preserving transformations fails, structural attribution is blocked, irrespective of how impressive or consistent the manifested behaviour may otherwise appear within an interaction regime.

8. Consequences for Evaluation and Ontology

The admissibility framework yields consequences that are primarily inferential and classificatory rather than empirical. Once the attribution gap is taken seriously, and once invariance under admissible identity-preserving transformations is installed as a necessary condition for structural status, a number of common moves in AI discourse become logically blocked. The consequences are not "claims about systems" in the performance sense, but constraints on what evaluators, institutions, and theorists are permitted to infer from performance evidence, and on how AI systems can be placed within ontological categories without committing a category error.

First, performance is rendered ontologically insufficient by construction. No accumulation of correct responses, no breadth of task completion, and no degree of behavioural regularity can, on its own, entail structural attribution, because such evidence remains confined to the observational domain and remains compatible with multiple generative explanations that do not instantiate the property as an identity-bearing feature of the system. Performance data can support the claim that P is manifestable, and may be indispensable for engineering and deployment decisions, but it cannot, without transformation-sensitive constraints, do the inferential work required for $P(S)$.

Second, the framework explains why mimicry of structure is not an anomaly but an expected compatibility class. Systems can behave as though they possess stable internal properties while in fact relying on configuration-bound mechanisms, including memorisation-like reuse of patterns, interaction residue, and other state-dependent processes that yield convincing manifestations without providing invariance across Σ_{state}. This compatibility is precisely what makes structural conservatism rational within the framework: the cost of false-positive ontology is high because it converts interaction-conditioned appearance into system identity without a valid bridge.

Third, these constraints reorganize ontological categorization. Predicates such as understanding, agency, and epistemic authority are treated as structurally heavyweight, meaning that their admissible attribution presupposes invariance under T. Where this invariance burden cannot be met, the appropriate ontological classification shifts away from epistemic subject-language and toward instrumental extension language: systems are treated as powerful mediating instruments embedded in human meaning-bearing structures, rather than as autonomous bearers of understanding. This classification is developed consequentially in Hansen (2026), *Artificial Intelligence as Instrumental Extension*, which links attribution discipline to governance-level implications for responsibility, oversight, and epistemic authority.

Fourth, evaluation itself is reframed. The question ceases to be "how well does the system complete tasks" and becomes "how stable is the manifested property under identity-preserving transformation." The evaluation target moves from aggregated task success to transformation-sensitive interrogation, where invariance violations function as decisive witnesses that block structural attribution. This is not a proposal to replace benchmarking for engineering purposes, but a proposal to prevent benchmarking results from being misread as ontological proof. Under the framework, evaluation that aims at ontology must be explicitly designed to test stability under admissible transformations, because without such tests, observation remains trapped on the wrong side of the attribution gap.

Taken together, these consequences establish the paper's practical philosophical claim: structural attribution is a gated regime, not an extrapolation from performance. The framework therefore does not tell us which systems are "smart," but it tells us what must be shown, and what cannot be inferred, before we are entitled to treat behavioural manifestation as evidence of constitutive understanding or subject-like ontology.

8.1 Performance Insufficiency

The admissibility framework treats performance evidence, even when extensive, as categorically incapable of *entailing*structural attribution. Accumulating correct responses, higher success rates, broader task coverage, or increasingly consistent behavioural regularities can strengthen a claim of the form "the system manifests P under the interaction regime I," but none of these accumulations, by themselves, can logically force the conclusion "P is instantiated in the system as such." The reason is not that performance data are uninformative, the reason is that they remain confined to the observational domain, and they therefore remain compatible with multiple underlying generative explanations that do not confer the property as an identity-bearing feature of the system.

This is the strict sense in which performance is insufficient. Performance evidence is, at best, *support* for output-level manifestation claims, and at most *suggestive* with respect to structure, but it lacks a deductive bridge to ontology because the mapping from constitutive organisation to observed behaviour is many-to-one. The same apparent correctness profile can arise from transient state alignment, context saturation, prompt scaffolding, distributional shortcuts, evaluator-side constraints, or other interaction-coupled mechanisms that do not amount to the kind of state-space stability that structural attribution presupposes. Consequently, no finite accumulation of correct outputs can eliminate the possibility that the observed property is configuration-bound rather than constitutive, unless additional admissibility constraints are introduced that explicitly test invariance under identity-preserving transformations.

The framework therefore refuses the implicit escalation that is common in evaluation discourse, where "enough" correct behaviour is treated as ontological authorization. Structural attribution is not a reward for repeated success. It is a claim with a different logical target, and it can become admissible only when the necessary invariance burden is met. In this sense, performance can accumulate without ever crossing the boundary into ontology, because what is missing is not quantity of correct responses, but a principled demonstration that the property survives relocation within Σ_{state} under the admissible transformation set T.

8.2 Memorisation Compatibility

A central reason performance cannot entail ontology is that behavioural success is compatible with mechanisms that mimic structural properties without instantiating the invariance conditions those properties require. Memorisation, in the broad sense of storing or reusing patterns in ways that do not produce identity-bearing stability, is a paradigmatic case. A system can produce outputs that appear to reflect stable understanding, rule possession, or internally grounded concepts while in fact operating through mechanisms that are configuration-bound, history-dependent, or interaction-scaffolded. In such cases the system can *simulate the surface signatures* of structure while lacking the state-space invariance that would justify treating the property as a system-level attribute.

The key point is not that memorisation is rare or pathological, but that it is structurally ambiguous with respect to ontology. A system may answer correctly because it has retrieved a pattern, matched a template, or activated a familiar representation that fits the current context, and it may do so with striking consistency within a given interaction regime. Yet this consistency can remain contingent on the system being held in a favourable configuration, for example through sustained context, repeated framing, or interaction sequences that keep the system within a narrow region of Σ_{state}. The property then appears stable in outputs, and the temptation to attribute it structurally becomes strong, but the stability is not an identity condition, it is an artefact of how the system has been driven, and it can collapse under identity-preserving transformations that disrupt transient residue.

This is why memorisation is described here as *compatible* with the appearance of structure. It provides a generative route to behaviour that can satisfy demanding property descriptions without requiring that the property be anchored in constitutive organisation. Under the invariance principle, such behaviour is insufficient for structural attribution precisely because it does not rule out state dependence. A memorisation-compatible mechanism can yield fluent explanations, consistent answers, and even coherent self-reference, while still failing the reset-based and transformation-based stability constraints that structural status entails.

Accordingly, the framework treats "mimicry of structure" as a predictable outcome of systems whose outputs are shaped by large stores of prior regularities and by interaction-sensitive configuration dynamics. The admissibility question is not whether such systems can produce convincing manifestations of P, but whether P survives identity-preserving relocation within the system's state domain. If it does not, then the system's behaviour remains a performance phenomenon, however impressive, and structural attribution remains inadmissible.

8.3 Ontological Categories

Once invariance is treated as a necessary condition for admissible structural attribution, the classification problem for AI systems shifts from a capability-centred vocabulary to an ontological one. The central distinction is no longer between "weak" and "strong" performance, but between systems whose observed properties remain interaction-indexed manifestations and systems for which structural attribution is admissible, meaning that at least some properties can coherently be treated as identity-bearing under admissible identity-preserving transformations. This reorientation has immediate consequences for how AI is placed within broader ontological categories, particularly the recurrent tendency to slide from behavioural competence into the language of agency, understanding, and epistemic subjecthood.

Under the framework developed here, the category *epistemic subject* is not a rhetorical elevation granted by fluency or correctness, but a classification that presupposes admissible system-level attribution of epistemically heavyweight predicates, most notably structural understanding $U(S)$. If $U(S)$ is constrained by invariance under T, then any system for which understanding-like behaviour fails persistence under identity-preserving transformations cannot be licensed as an epistemic subject in the relevant sense, regardless of how consistently it manifests the surface signatures of understanding within an interaction regime. The point is not that such a system is unintelligent "in practice," but that the ontological predicate being invoked requires a stability burden that performance alone cannot supply. In that case, subject-language becomes an attributional overreach: it treats a property of outputs, or of transient configurations, as though it were an identity condition of the system.

By contrast, the category *instrumental extension* becomes the natural ontological placement for systems whose properties are performance-manifested but not structurally admissible. A system can be extraordinarily useful, operationally reliable, and economically transformative while still remaining, in the strict ontological sense, an instrument embedded in and dependent upon human meaning-bearing structures, rather than a self-standing epistemic bearer. The framework provides a disciplined explanation of why this category is not a normative demotion but a logical classification: if the decisive predicates cannot be attributed structurally, then the system's role is best described as amplificatory and mediational, a tool that extends human cognition and action without thereby constituting a new epistemic subject-category.

This point is developed consequentially in Hansen (2026), *Artificial Intelligence as Instrumental Extension*, which treats the attribution discipline articulated here as a prerequisite for governance-level claims about agency, responsibility, and epistemic authority. The link is direct: once structural status is denied on invariance grounds, subject-like attributions become not merely uncertain but category-invalid within the admissibility regime, and the downstream consequences shift accordingly, from questions of "how smart is the system" to questions of how its outputs should be integrated, constrained, audited, and socially interpreted as instrumentally mediated artefacts rather than as the utterances of an autonomous epistemic agent.

8.4 Evaluation Reframing

The invariance principle implies a reframing of evaluation that is methodological rather than rhetorical. If structural attribution is governed by stability under admissible identity-preserving transformations, then the evaluative centre of gravity cannot remain task completion alone. Task completion measures whether a property appears in outputs under a chosen interaction regime, but it does not test whether the property is an identity-bearing feature of the system, because it typically leaves the system within a narrow and often unexamined band of Σ_{state}, shaped by the very interaction history that makes success likely.

Under the admissibility framework, the relevant evaluative question therefore becomes: does the property persist when the system is relocated within its admissible state space through transformations that preserve identity. This shifts emphasis from "can the system produce correct outputs" to "does the property survive normalization, reset, context perturbation, and other admissible transformations $\tau \in T$ that should not matter if the property is truly structural." The object of evaluation changes accordingly. Instead of aggregating success rates across tasks, the evaluator must confront invariance as a boundary condition, seeking counterexamples under identity-preserving relocation rather than accumulating confirmations within a fixed regime.

This reframing also clarifies why the framework is not merely proposing a different kind of benchmark. It is altering the inferential target of evaluation. In a task-completion paradigm, the natural endpoint is a scalar score that summarizes performance. In an invariance paradigm, the natural endpoint is an admissibility judgement: whether the conditions for system-level attribution remain intact under transformation, or whether the property collapses into state dependence. The framework does not deny the utility of task evaluation for engineering purposes, but it denies its ontological sufficiency. When the question is whether $P(S)$ is admissible, the decisive evidence is not how often $P(O)$ appears, but whether P can be made to fail under identity-preserving transformation, thereby revealing that the apparent property was configuration-bound rather than constitutive.

In this sense, evaluation is reframed from performance aggregation to stability interrogation. The normative implication is not that tasks are irrelevant, but that tasks must be embedded within transformation-sensitive protocols if they are to bear on structural attribution. The framework therefore relocates the burden of proof: the path to admissible ontology runs through invariance under T, not through the accumulation of correct answers under a fixed interaction regime.

9. Conclusion

The argument of the paper has been a disciplined restriction on what can be inferred from observation, rather than an expansion of what can be claimed on the basis of performance. It begins from the non-identity premise $P(O) \neq P(S)$, which blocks the default escalation from manifested properties in output to instantiated properties in system constitution. Observational regularity, even when strong and broadly quantified over an interaction regime, remains indexed to the wrong domain to warrant ontological conclusions, because behavioural evidence underdetermines internal organisation and is compatible with multiple generative explanations, including configuration-bound and scaffold-dependent ones.

From this starting point, the paper's central contribution is the formal clarification of admissibility conditions for structural attribution. By separating the performance and structural regimes through explicit quantification domains, and by introducing a minimal binary admissibility function constrained by necessary requirements and asymmetrical falsification, the framework makes ontological language a gated regime rather than a reward for repeated task success. The reset operator R and the admissible transformation set T operationalize the missing bridge identified in the observational problem: they provide a principled way to test whether an apparent property survives identity-preserving normalization and relocation within Σ_{state}. This yields the paper's core thesis as a necessary condition, namely that structural properties, and especially structural understanding $U(S)$, require invariance under identity-preserving transformation.

The limits of the paper are explicit and non-negotiable. No empirical sufficiency claims are made for any specific system. The paper does not evaluate performance, compare architectures, or propose a benchmark. Its scope is confined to the logic of admissible attribution, and its conclusions are constraints on inference and classification, not verdicts about capability. In that sense, the paper is best read as a conceptual control mechanism for AI discourse and governance: it specifies what must be shown before system-level predicates, including understanding and subject-like ontology, can be treated as warranted rather than as behaviourally induced projection.

9.1 Summary

The paper's argument begins from a disciplined refusal of the default inference from output manifestation to system instantiation. The core premise is the non-identity of observational and structural property claims: $P(O) \neq P(S)$. Observing a property in outputs, even with strong behavioural regularity, establishes at most that the property is manifestable under an interaction regime, not that it is an identity-bearing feature of the system as such.

From this non-identity premise follows the central necessary-condition thesis: structural attribution is admissible only under an invariance burden. If a property is attributed to the system, rather than merely to its outputs or transient configurations, then it must persist under admissible identity-preserving transformations that relocate the system within Σ_{state} while holding parameters, code and architecture, and constitutive external scaffolding constant. The reset operator and the broader transformation set T formalize this requirement as a methodological bridge-test: where invariance fails, structural status is denied, not as a claim about system capacity, but as a constraint on admissible ontology.

9.2 Contribution

The contribution of the paper is a formal clarification of the conditions under which structural attribution is admissible, not an empirical evaluation of any particular system. It isolates the attribution gap as a domain mismatch between interaction-quantified observation and state-space-quantified constitution, and it hardens invariance under admissible identity-preserving transformations as a necessary constraint for treating properties as system-level rather than output-level. By introducing a minimal binary admissibility regime, an explicit requirement predicate, and a reset-based diagnostic bridge-test, the paper provides a disciplined vocabulary and logical structure for preventing the common escalation from performance to ontology. The result is not a new benchmark or a comparative claim, but a rule-governed framework for determining when ontological language is warranted and when it is categorically inadmissible.

9.3 Final Thesis

The final thesis is therefore a necessary-condition claim about what it means to attribute understanding structurally. Structural understanding is an invariance condition: to assert $U(S)$ as a system-level property is to accept the burden that understanding must persist under admissible identity-preserving transformations T across the system's admissible state domain Σ_{state}. Where understanding-like behaviour fails under such transformations, the correct classification is performance-level manifestation or state-dependent configuration effect, not constitutive understanding.

The paper makes no claim of empirical sufficiency for any specific system. It does not evaluate particular architectures, report benchmark outcomes, or assert that any existing system satisfies the admissibility conditions developed herein. Its analysis is confined to the logical requirements for admissible attribution, and its conclusions should be read as constraints on inference and classification, not as empirical verdicts about concrete systems.

Appendix A - Formal Lemma and Extended Proof

A.1 Proposition

This appendix states a strengthened version of the invariance claim as a biconditional, explicitly treating invariance not only as a necessary condition for structural attribution, but as a sufficient one within the paper's *stipulated* admissibility regime. The formulation is placed in the appendix precisely because, in the main argument, the paper hardens invariance as necessary to avoid the appearance of definitional circularity. Here, by contrast, the intent is to make explicit what one may adopt as a *closure rule* for classification once the identity-preserving transformation set T has been fixed and justified.

$$P(S) \Leftrightarrow \forall\tau \in T : P(S_\tau)$$

Read strictly, the biconditional should be interpreted as *relative sufficiency*: if T is adequately specified as the set of admissible identity-preserving transformations, and if "identity preservation" has been hardened so that parameterization, code and architecture, and constitutive external scaffolding are constant while transient state is the only locus of variation, then invariance under T is taken to be sufficient for treating P as a structural property of S. The sufficiency claim is therefore not metaphysical, it is methodological: it asserts that, *given the chosen notion of identity and admissibility*, no further bridge is required beyond invariance under the sanctioned transformation regime.

Two constraints follow immediately.

1. The strength of the biconditional is only as good as the adequacy of T. If T is too weak, the "sufficient" direction becomes permissive and risks re-admitting configuration-bound properties as structural; if T is too strong, the criterion becomes unrealizable and collapses structural attribution into vacuity.
2. The biconditional is a governance-grade closure rule, not an empirical claim about how properties arise in real systems. It specifies when attribution is *licensed* under the framework, not how a system internally produces the stability that satisfies the license.

Accordingly, the main text treats invariance as a necessary condition to preserve non-circular argument structure, while this appendix records the optional, explicitly stipulated equivalence that one may adopt when the aim is a crisp classification rule rather than an inferentially minimal constraint.

A.2 Proof Sketch

The proposition in A.1 can be justified as a biconditional once two commitments are made explicit: first, that the transformation class T is *identity-preserving* in the hardened sense adopted in Section 5, meaning that parameters, code and architecture, and constitutive external scaffolding are held constant while only transient configuration is allowed to vary, and second, that T is *admissibility-adequate*, meaning that it is rich enough to expose any state-dependence that would otherwise permit output-level mimicry to be mistaken for system-level instantiation.

Necessity $(\boldsymbol{P}(\boldsymbol{S}) \Rightarrow \forall\boldsymbol{\tau} \in \boldsymbol{T}\colon \boldsymbol{P}(\boldsymbol{S_\tau}))$

Assume $P(S)$, where P is intended as a **structural** attribution, that is, a claim about the system *as such* rather than about an interaction episode or a transient configuration.

1. By definition of identity-preservation, every $\tau \in T$ produces an instantiation S_τ that differs from S only in transient configuration, not in identity conditions.
2. A structural attribution is, by its semantic content, an identity-level claim, meaning it cannot coherently be hostage to variation that leaves identity fixed.
3. Therefore, for each admissible $\tau \in T$, the attributed property must remain satisfied after transformation, yielding $P(S_\tau)$.
4. Since τ was arbitrary, $\forall \tau \in T: P(S_\tau)$ follows.

This direction is essentially the formal version of the invariance principle: if a property is truly attributed to the system in virtue of constitutive organisation, then it must survive any transformation that leaves that constitution unchanged.

Sufficiency ($\forall \boldsymbol{\tau} \in \boldsymbol{T}: \boldsymbol{P}(\boldsymbol{S}_{\boldsymbol{\tau}}) \Rightarrow \boldsymbol{P}(\boldsymbol{S})$)

The reverse direction is where the "under defined transformation classes" clause does real work. The implication is not metaphysically automatic, it is secured by how T is specified and by what is meant by "structural" within the admissibility regime.

Assume $\forall \tau \in T: P(S_\tau)$. Two steps establish sufficiency.

Step 1: Recover the baseline instance.
Because T is a class of admissible transformations, it includes (or is assumed to be compatible with) the identity transformation id, or else a designated canonicalization mapping that yields the baseline instantiation. Hence:

$$P(S_{id}) \text{ holds, and } S_{id} = S \Rightarrow P(S).$$

This already yields the conclusion at the level of the symbol S, but it does not yet justify *structural* status in the intended sense unless T is adequate to test state-dependence.

Step 2: Eliminate state-dependence by coverage.
Introduce the adequacy condition that connects the transformation class to the state-space domain: for any admissible internal state $\sigma \in \Sigma_{\text{state}}$, there exists a transformation $\tau_\sigma \in T$ that realizes that state (or reaches a representative of its equivalence class) from a fixed baseline $\sigma_0 \in \Sigma_{\text{base}}$:

$$\forall \sigma \in \Sigma_{state}\ \exists \tau_\sigma \in T : \tau_\sigma(\sigma_0) = \sigma.$$

Given $\forall \tau \in T: P(S_\tau)$, we obtain in particular that $P(S_{\tau_\sigma})$ holds for every σ, which entails that P holds across the entire admissible state regime, not merely for a favourable configuration. Under the regime where "structural" is taken to mean "state-space stable under identity-preserving relocation," this is precisely what licenses structural attribution. In other words, the invariance hypothesis, coupled with coverage, rules out the configuration-bound explanations that motivated the admissibility framework in the first place.

Scope of the Sketch

The biconditional is therefore valid **relative to the specified transformation regime**. The necessity direction depends only on identity-preservation plus the intended semantics of structural attribution.

The sufficiency direction depends on the additional adequacy requirement that T is strong enough, by design, to traverse or represent the relevant variation in Σ_{state}. If T is under-specified, for example if it excludes transformations that would reveal dependence on context residue, caching, or other transient configuration variables, then $\forall \tau \in T: P(S_\tau)$ can hold while P remains non-structural in the stronger intended sense. This is why the main text treats invariance as necessary, while the appendix records the strengthened equivalence as a closure rule available once T has been explicitly justified as admissibility adequate.

A.3 Justification

The sufficiency direction in A.1 is not asserted as a metaphysical fact about properties in general, but as a *closure rule internal to the admissibility regime* once the transformation class T has been fixed under hardened identity-preservation constraints and declared adequate for the purpose of distinguishing structural instantiation from configuration-bound manifestation. This framing is what prevents the biconditional from collapsing into circularity.

The potential circularity charge typically takes the form: "You have defined 'structural' to mean 'invariant', and then you conclude that invariance implies structurality." The present appendix avoids that structure by making explicit that the equivalence is conditional on a prior, independently motivated construction of T. The paper's main argument motivates the need for a bridge from observation to ontology on grounds of domain asymmetry and underdetermination, and it motivates the need for identity-preserving transformations on grounds of what it means to attribute a property to the system *as such* rather than to an interaction trajectory. From that motivation, T is not introduced as a definitional artifact designed to force a conclusion, but as the minimal methodological apparatus required to test whether an apparent property survives relocation within Σ_{state} while system identity is held fixed.

Given such a T, sufficiency is justified in the following precise sense: if a property persists across all admissible identity-preserving transformations in T, then, relative to the regime's own account of what counts as "identity-bearing," no further evidence is required to license system-level attribution. Invariance becomes sufficient because the regime has already declared that the only admissible variation relevant to the distinction between manifestation and instantiation is the variation represented by T. Put differently, sufficiency is not "proved" by definition, it is *earned* by the prior hardening of identity preservation and by the declared adequacy of T as a transformation set that, by design, targets precisely the failure modes that make observational universality insufficient, namely state dependence, context residue, and other transient configuration effects.

This is why the biconditional is placed in the appendix rather than the main text. The main argument retains invariance as a necessary condition to keep the core inferential posture non-stipulative and to avoid any impression that the framework is a definitional loop. The appendix then records an optional strengthening: once a community, a governance regime, or a methodological protocol has explicitly committed to a particular T as the sanctioned identity-preserving transformation class, it may adopt invariance under that T as a sufficient licensing criterion for structural attribution. The sufficiency claim is therefore explicitly relative, transparently conditional, and thereby insulated against circularity objections, because the work is carried by the prior specification and justification of T, not by a covert redefinition of "structure" in terms of the conclusion the framework seeks to reach.

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